

MS 120 Vector Space Extra Problems

September 20, 2026

1 Worked examples

Here are 3 worked examples for you to model after for homework.

Make sure you can work these by hand on the **test**. I will re-typeset these questions as they will appear on the test below in section 2.

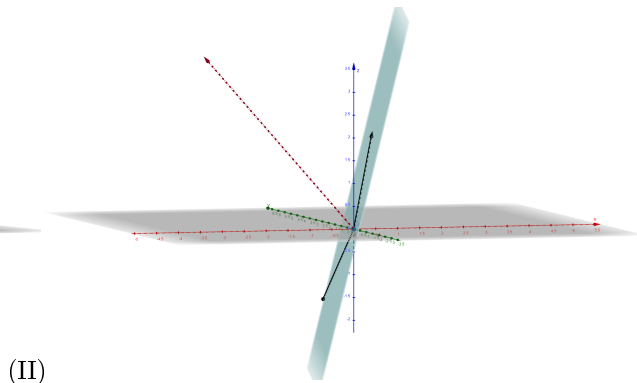
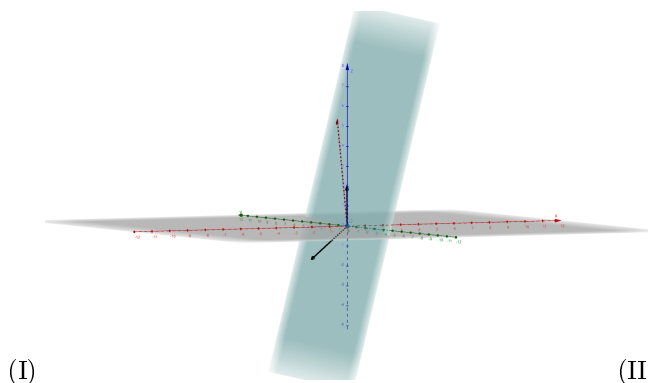
I will work the examples in Julia and upload the Julia file to my GitHub page.

It will be called `MS_120-vector-space-extra-problems.jl`.

1. Given

$$\begin{cases} 6x + 2y + z = 2 \\ y - z = -1 \\ -x - y - 3z = -2 \end{cases}$$

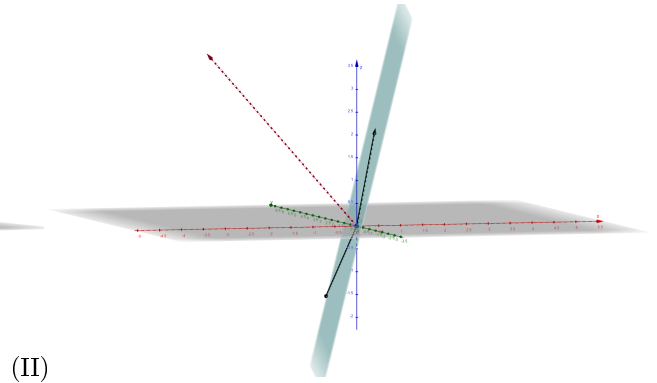
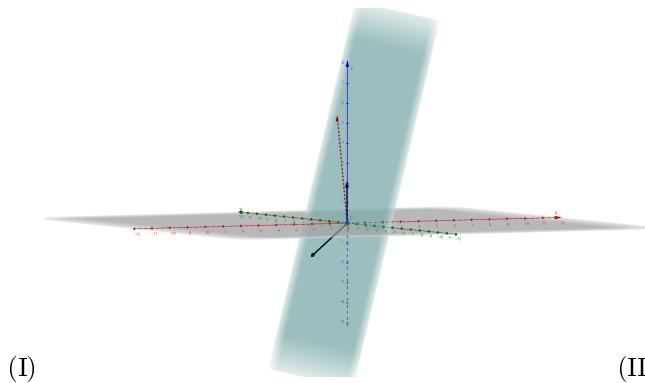
- (a) Rewrite the system of equations in the form $Ax = b$.
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- (e) Rewrite the matrix as column vectors.
- (f) Do the column vectors form a basis for the space that is being mapped into?
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- (i) If the dimension of the column space is not the maximum size possible, determine if b is in the column space or not. (Why or why not?) Which diagram best represents the geometry?



2. Given

$$\begin{cases} -x + 2y + 5z = 2 \\ x - 2z = 0 \\ 4x - 2y - 11z = 7 \end{cases}$$

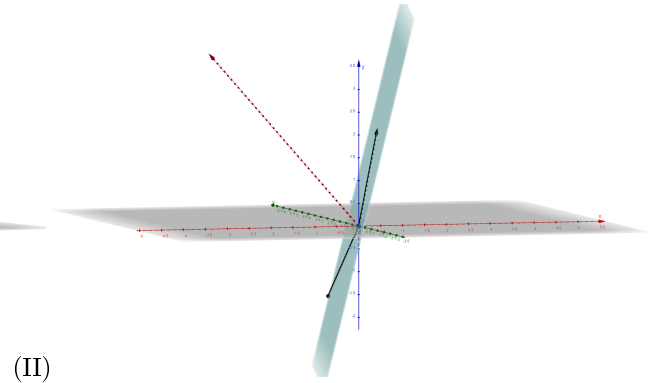
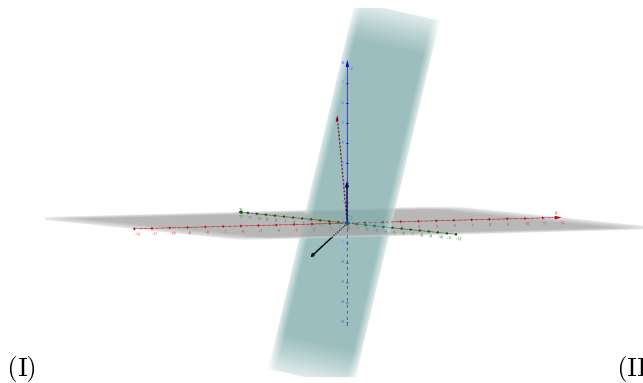
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- (i) If the dimension of the column space is not the maximum size possible, determine if b is in the column space or not. (Why or why not?) Which diagram best represents the geometry?



3. Given

$$\begin{cases} x - 2y + z = 3 \\ 2x - 5y + 6z = 7 \\ 2x - 3y - 2z = 5 \end{cases}$$

- (a) Rewrite the system of equations in the form $Ax = b$.
- (b) What is the size of the matrix A ?
- (c) What is the dimension of the space that is being mapped:
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2 As seen on test.

1. Given

$$\begin{cases} 6x + 2y + z = 2 \\ y - z = -1 \\ -x - y - 3z = -2 \end{cases}$$

(a) Rewrite the system of equations in the form $Ax = b$.

(b) What is the size of the matrix A ?

(c) What is the dimension of the space that is being mapped:

(I) from?

(II) to?

(d) What is the maximum size the column space could be given the size of the matrix?

(e) Rewrite the matrix as column vectors.

(f) Do the column vectors form a basis for the space that is being mapped into?

```
julia> det(A)
-21.0
```

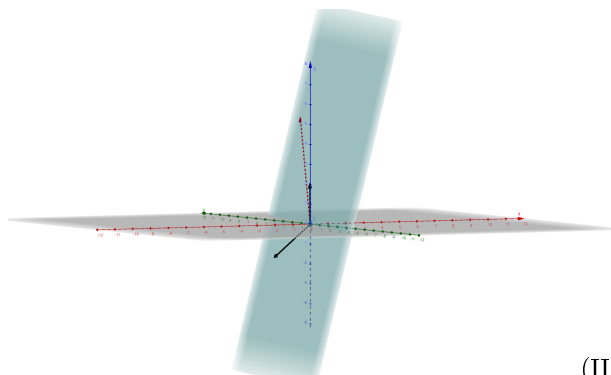
(g) What is the dimension of the column space?

(h) If the dimension of the column space is the maximum size possible is b in the column space?

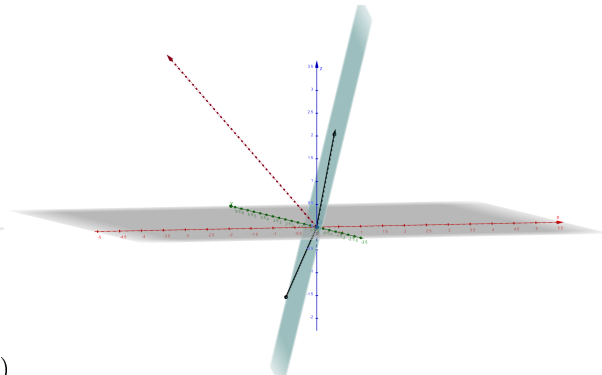
(i) If the dimension of the column space is not the maximum size possible, determine if b is in the column space or not. (Why or why not?) Which diagram best represents the geometry?

```
julia> rref([A b])
3×4 Matrix{Float64}:
 1.0  0.0  0.0  0.333333
 0.0  1.0  0.0 -0.333333
 0.0  0.0  1.0  0.666667
```

(I)



(II)



2. Given

$$\begin{cases} -x + 2y + 5z = 2 \\ x - 2z = 0 \\ 4x - 2y - 11z = 7 \end{cases}$$

(a) Rewrite the system of equations in the form $Ax = b$.

(b) What is the size of the matrix A ?

(c) What is the dimension of the space that is being mapped:

(I) from?

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(f) Do the column vectors form a basis for the space that is being mapped into?

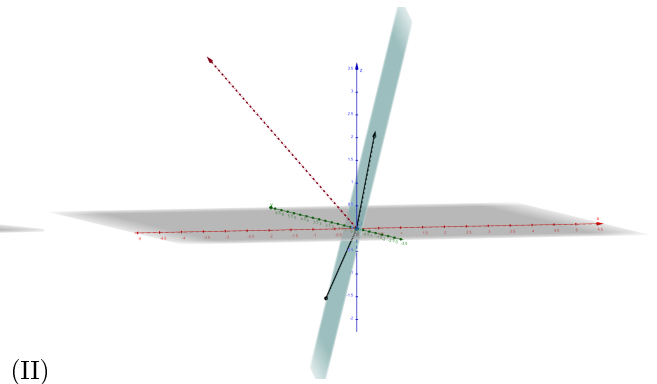
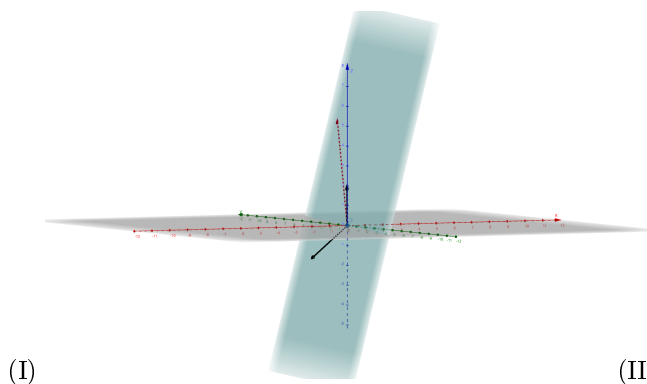
```
julia> det(A)
0.0
```

(g) What is the dimension of the column space?

(h) If the dimension of the column space is the maximum size possible, is b not in the column space?

(i) If the dimension of the column space is not the maximum size possible, determine if b is in the column space or not. (Why or why not?) Which diagram best represents the geometry?

```
julia> rref([A b])
3×4 Matrix{Float64}:
1.0  0.0 -2.0  0.0
0.0  1.0  1.5  0.0
0.0  0.0  0.0  1.0
```



3. Given

$$\begin{cases} x - 2y + z = 3 \\ 2x - 5y + 6z = 7 \\ 2x - 3y - 2z = 5 \end{cases}$$

(a) Rewrite the system of equations in the form $Ax = b$.

(b) What is the size of the matrix A ?

(c) What is the dimension of the space that is being mapped:

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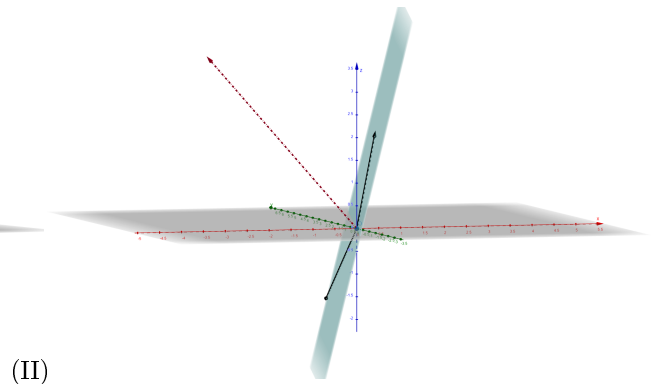
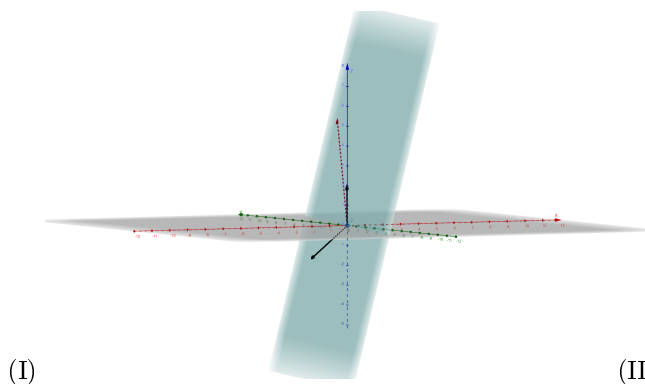
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julia> det(A)
0.0
```

(g) What is the dimension of the column space?

(h) If the dimension of the column space is the maximum size possible is b in the column space?

(i) If the dimension of the column space is not the maximum size possible, determine if b is in the column space or not. (Why or why not?) Which diagram best represents the geometry?

```
julia> rref([A b])
3×4 Matrix{Float64}:
1.0  0.0 -7.0  1.0
0.0  1.0 -4.0 -1.0
0.0  0.0  0.0  0.0
```

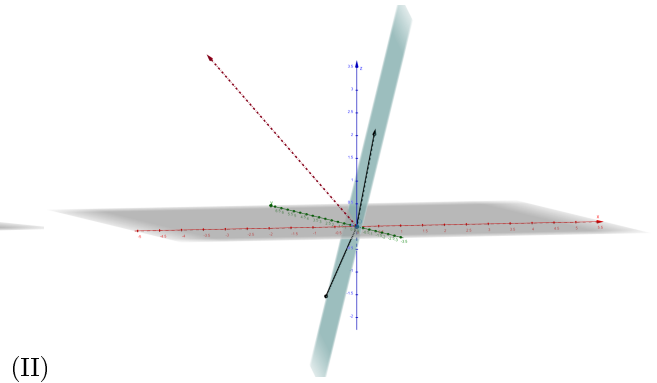
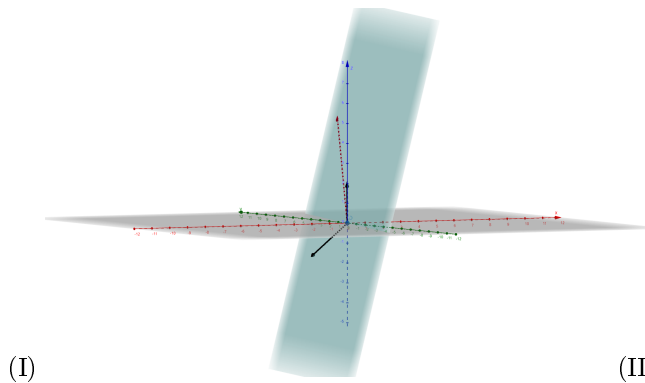


3 Homework.

1. Given

$$\begin{cases} x - 3y = 14 \\ 3x + 2y = -13 \end{cases}$$

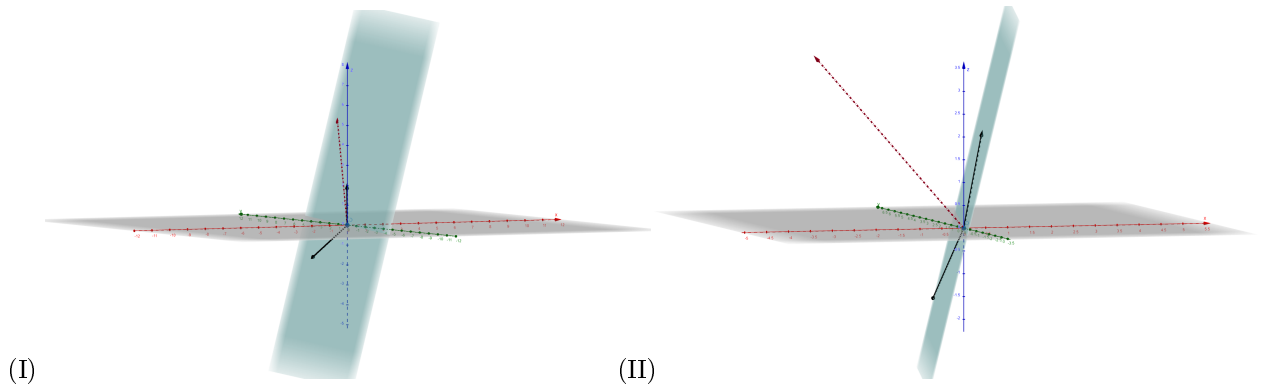
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2. Given

$$\begin{cases} x - y = 5 \\ 2x + y = 19 \end{cases}$$

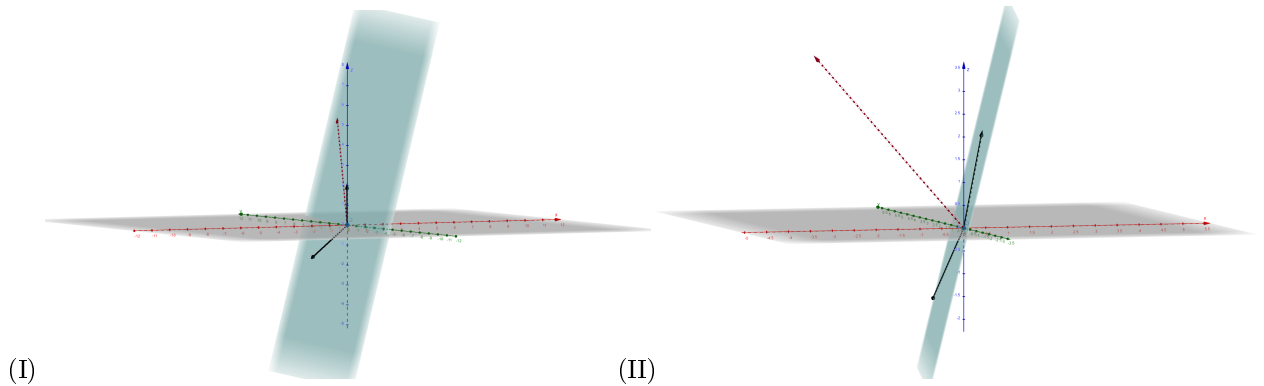
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3. Given

$$\begin{cases} 3x + 9y = -15 \\ x + 3y = 6 \end{cases}$$

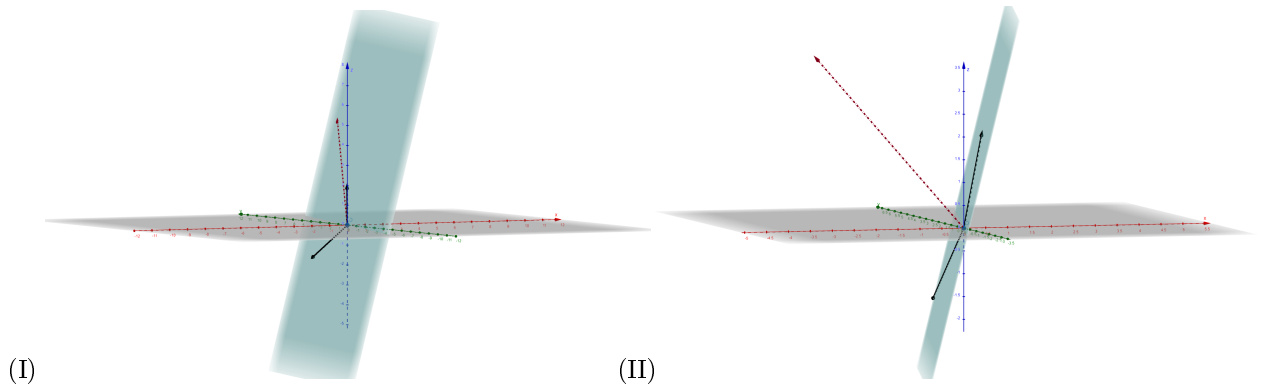
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4. Given

$$\begin{cases} 3x - y = 6 \\ 6x - 2y = 18 \end{cases}$$

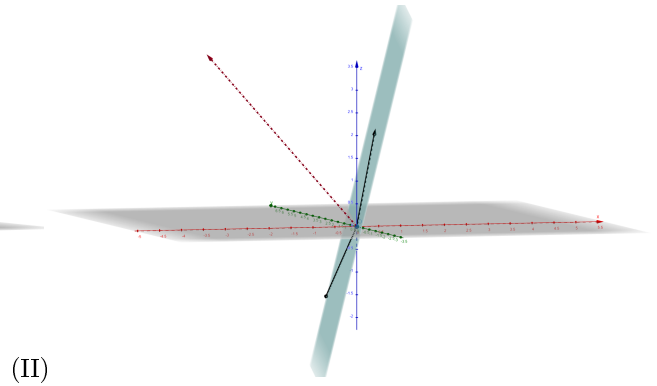
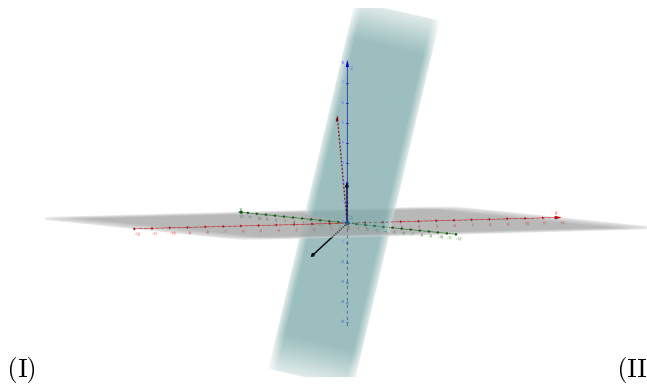
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5. Given

$$\begin{cases} 4x - 9y = -60 \\ \frac{1}{3}x + \frac{3}{4}y = -5 \end{cases}$$

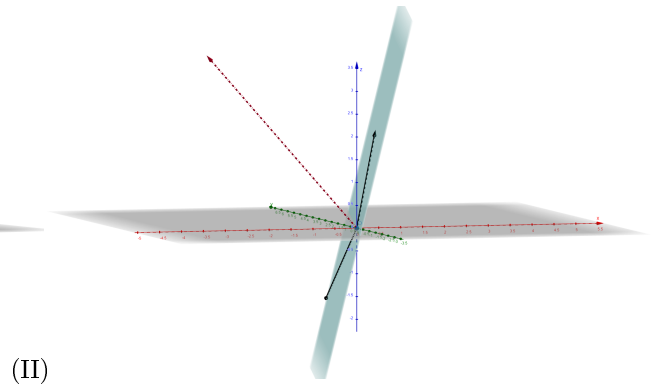
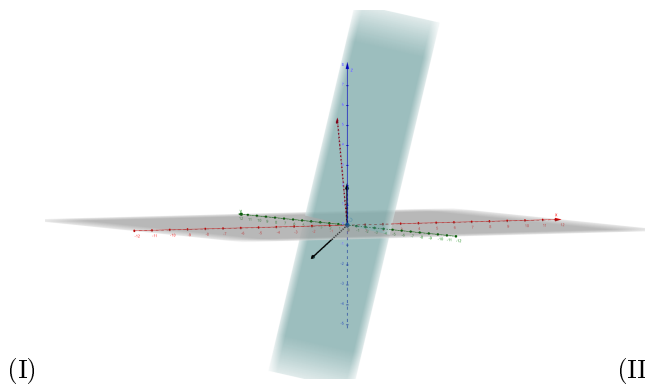
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6. Given

$$\begin{cases} 2x - 3y + 4z = 3 \\ 5y - 2z = -19 \\ 3z = 6 \end{cases}$$

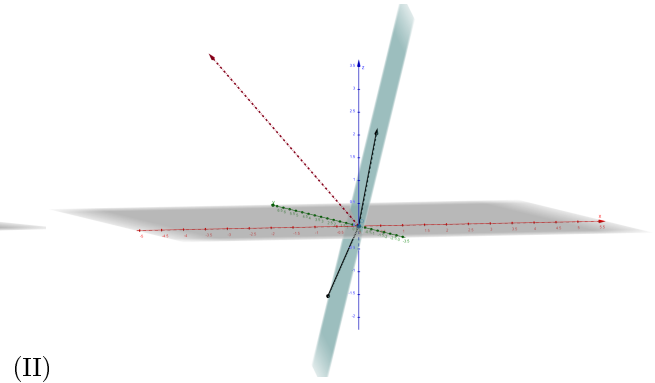
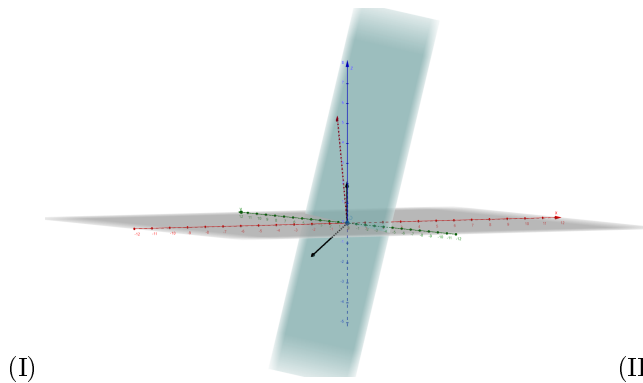
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7. Given

$$\begin{cases} x - 2y + 3z = 20 \\ 2x + y + 5z = 40 \\ 3x - 4y - 2z = 7 \end{cases}$$

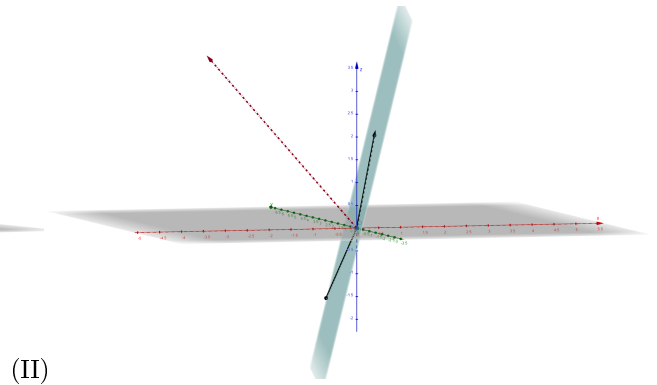
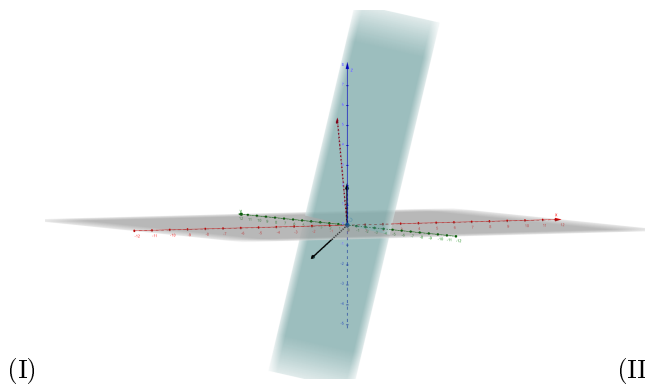
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8. Given

$$\begin{cases} x - 2y + z = -5 \\ 2x + 4y - 3z = -3 \\ -3x - 6y + 7z = 7 \end{cases}$$

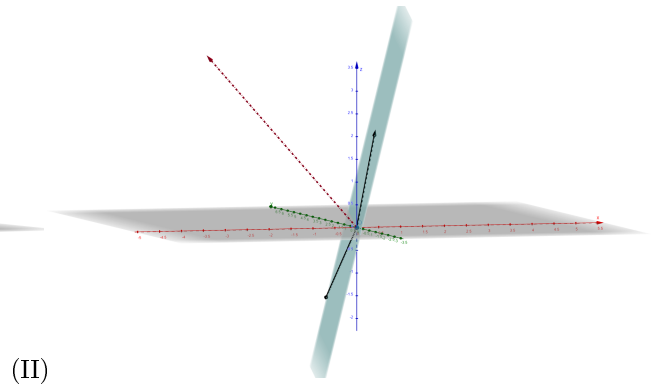
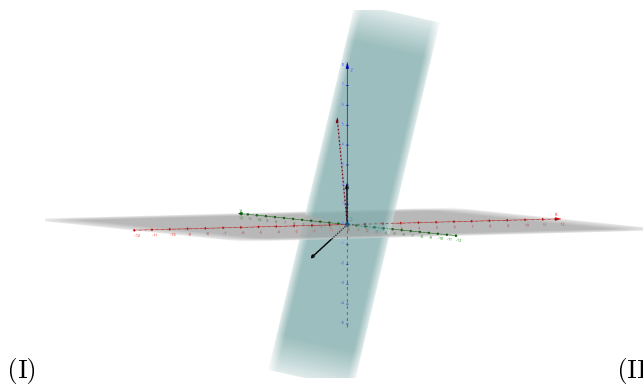
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9. Given

$$\begin{cases} x - 3y - 2z + w = -6 \\ -2x + 7y + z - 2w = -1 \\ 3x - 7y - 3z + 3w = -8 \\ 5x + y + 4z - 2w = 24 \end{cases}$$

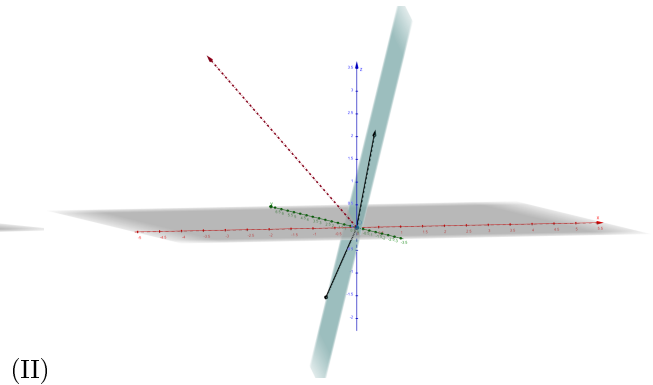
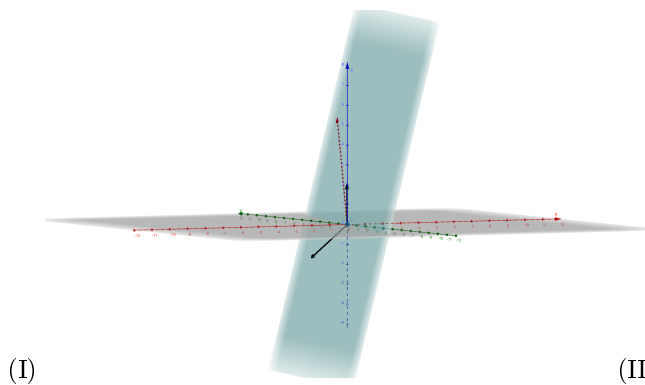
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- (i) If the dimension of the column space is not the maximum size possible, determine if b is in the column space or not. (Why or why not?) Which diagram best represents the geometry?



10. Given

$$\begin{cases} x + 2y - 3y + w = -8 \\ -2x - 3y + y - w = 14 \\ 4x + 9y - 2y - 2w = -55 \\ -5x - 9y + 2y - 3w = 41 \end{cases}$$

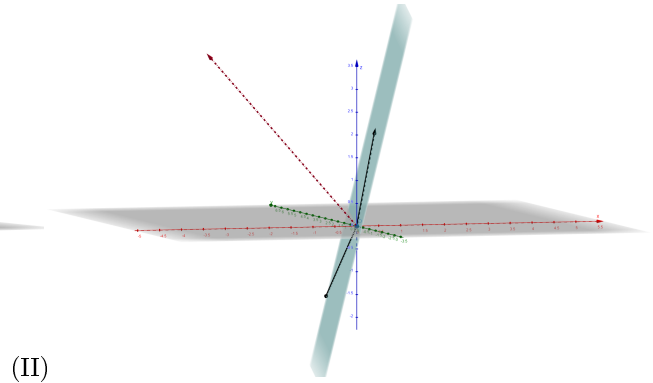
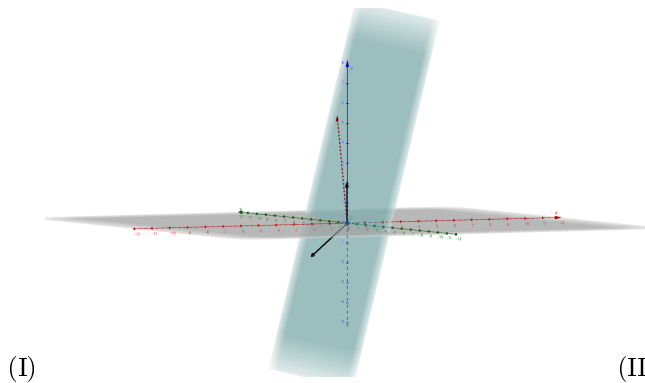
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11. Given

$$\begin{cases} x + 2y - z = 1 \\ 2x + 3y - 4z = -3 \\ 3x + 6y - 3z = 4 \end{cases}$$

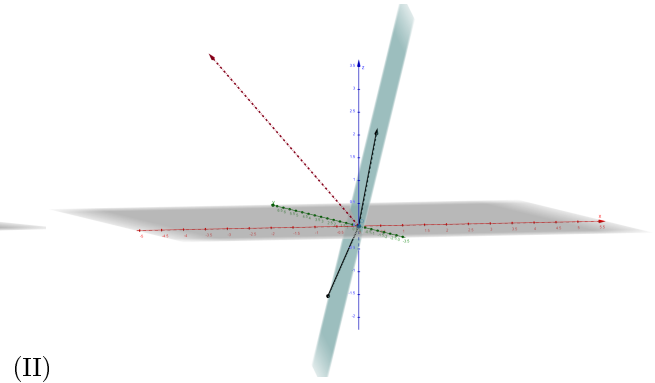
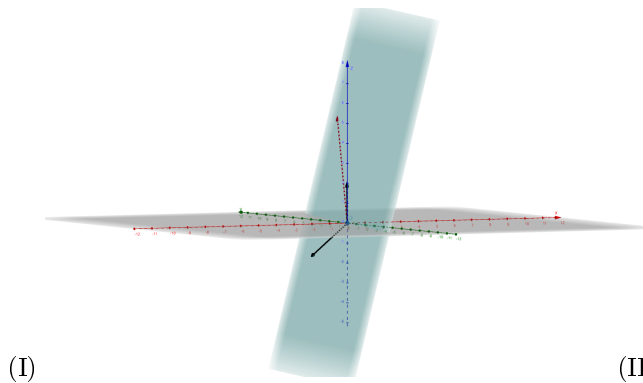
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12. Given

$$\begin{cases} x - 2y - 3z = 3 \\ 2x + y - z = 3 \\ 4x - 3y - 7z = 3 \end{cases}$$

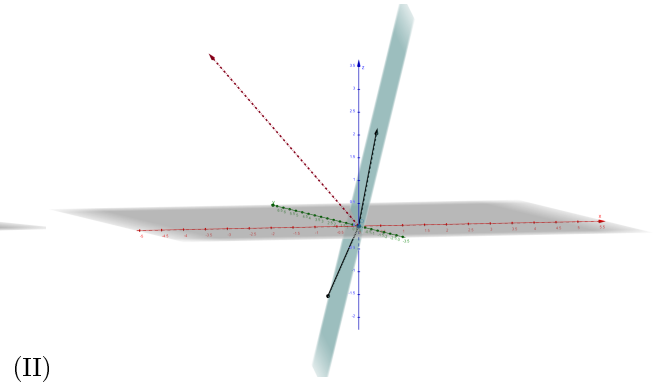
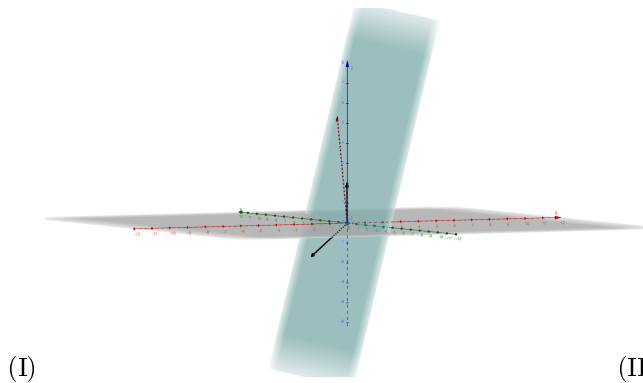
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- (b) What is the size of the matrix A ?
- (c) What is the dimension of the space that is being mapped:
 - (I) from? (II) to?
- (d) What is the maximum size the column space could be given the size of the matrix?
- (e) Rewrite the matrix as column vectors.
- (f) Do the column vectors form a basis for the space that is being mapped into?
- (g) What is the dimension of the column space?
- (h) If the dimension of the column space is the maximum size possible (on the test I will only use one of the following questions),
 - (I) is b in the column space?
 - (II) is b not in the column space?
- (i) If the dimension of the column space is not the maximum size possible, determine if b is in the column space or not. (Why or why not?) Which diagram best represents the geometry?



13. Given

$$\begin{cases} x + y - z = 0 \\ x + 2y - 3z = -2 \\ 2x + 3y - 4z = -2 \end{cases}$$

- (a) Rewrite the system of equations in the form $Ax = b$.
- (b) What is the size of the matrix A ?
- (c) What is the dimension of the space that is being mapped:
 - (I) from? (II) to?
- (d) What is the maximum size the column space could be given the size of the matrix?
- (e) Rewrite the matrix as column vectors.
- (f) Do the column vectors form a basis for the space that is being mapped into?
- (g) What is the dimension of the column space?
- (h) If the dimension of the column space is the maximum size possible (on the test I will only use one of the following questions),
 - (I) is b in the column space?
 - (II) is b not in the column space?
- (i) If the dimension of the column space is not the maximum size possible, determine if b is in the column space or not. (Why or why not?) Which diagram best represents the geometry?



14. Given

$$\begin{cases} x + 3y - 2z = 0 \\ 2x \quad \quad + 4z = 2 \\ 4x + 6y \quad \quad = 2 \end{cases}$$

- (a) Rewrite the system of equations in the form $Ax = b$.
- (b) What is the size of the matrix A ?
- (c) What is the dimension of the space that is being mapped:
 - (I) from? (II) to?
- (d) What is the maximum size the column space could be given the size of the matrix?
- (e) Rewrite the matrix as column vectors.
- (f) Do the column vectors form a basis for the space that is being mapped into?
- (g) What is the dimension of the column space?
- (h) If the dimension of the column space is the maximum size possible (on the test I will only use one of the following questions),
 - (I) is b in the column space?
 - (II) is b not in the column space?
- (i) If the dimension of the column space is not the maximum size possible, determine if b is in the column space or not. (Why or why not?) Which diagram best represents the geometry?

